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UV-curable hyperbranched poly(ester-co-vinyl) by radical ring-opening copolymerization for antifouling coatings†

Xiaoqing Ai, Jiansen Pan, Qingyi Xie, Chunfeng Ma * and Guangzhao Zhang*

Hyperbranched polymers with high solubility, low viscosity, low crystallinity and dense functional groups are promising to be used in adhesives and coatings. In this study, we have developed a cleavable hyperbranched poly(ester-co-vinyl) with diethylene glycol units and unreacted pendant vinyl groups via radical ring-opening copolymerization of 2-methylene-1,3-dioxepane (MDO), vinyl acetate (VAc) and diethylene glycol divinyl ether. Such a hyperbranched polymer with vinyl groups can form a crosslinked network by UV curing. The cleavable ester bonds are randomly distributed in the main chain, so they can degrade into small fragments with molar masses of 400–600 g mol⁻¹. The degradation rate can be well regulated with MDO content. The protein resistance, antibacterial adhesion and anti-diatom activity of the hyperbranched polymer were examined. The polymer with diethylene glycol units exhibits remarkable antifouling abilities and can effectively inhibit the adhesion of protein, marine bacteria (*Pseudomonas* sp.) and diatoms (*Navicula incerta*). The hyperbranched polymer coating can find application in marine antifouling.

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Introduction

Biodegradable polymers with biocompatibility and recyclability have found applications in tissue engineering,^{1–3} drug delivery,^{4–7} and coatings for medical devices.^{8,9} They were also used to prepare marine antibiofouling coatings in that the biodegradability makes them form a continuously changing surface in seawater, which can effectively inhibit the adhesion of marine biofouling.^{10–13} Generally, the biodegradation and mechanical properties can be regulated by the chemical composition and structure.

Recently, degradable vinyl polymers synthesized via radical ring-opening polymerization (rROP) of cyclic ketene acetals (CKAs) have attracted attention.^{14–16} Particularly, such polymerization of 2-methylene-1,3-dioxepane (MDO), a type of CKA, yields a polymer identical to poly(ϵ -caprolactone) (PCL).^{17,18} The monomer can copolymerize with a vast range of radically polymerizable vinyl monomers, such as methyl methacrylate (MMA),¹⁹ styrene (St),²⁰ *N*-vinylpyrrolidone (NVP),²¹ vinyl acetate (VAc),^{22,23} and vinyl ethers (VE).²⁴ The copolymer of MDO and MMA exhibits low crystallinity and improved mechanical properties.^{25,26} However, since the reactivity ratio of MDO ($r_{\text{MDO}} = 0.023$) is quite different from that of MMA

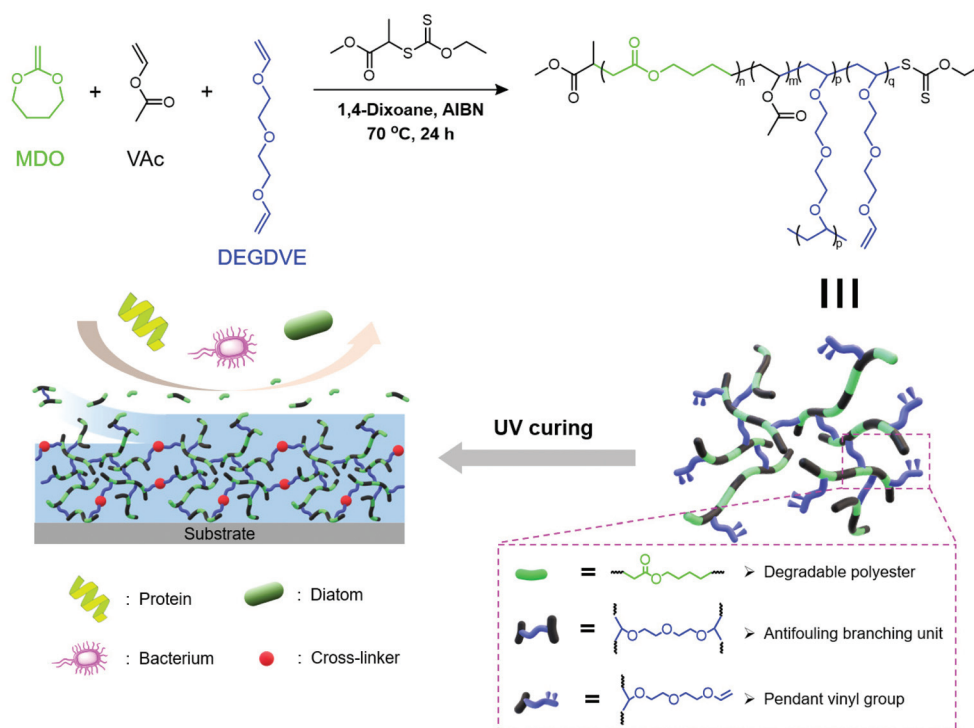
($r_{\text{MMA}} = 26.53$), the polymerization yields a block copolymer that exhibits a non-random degradation.²⁷ In the copolymerization of MDO with VAc ($r_{\text{MDO}} = 0.93$ and $r_{\text{VAc}} = 1.71$)²⁸ and VE ($r_{\text{MDO}} = 0.97$ and $r_{\text{VE}} = 0.99$),²⁴ the comonomers have a close reactivity ratio yielding a random copolymer chain that can randomly degrade into smaller fragments. Considering that they exhibit a low glass transition temperature, chemical modification is needed to improve their mechanical properties.

Hyperbranched polymers with unique chemical structures have dense end-up functional groups such as hydroxyl and vinyl groups which provide the possibility for further modification.^{29–33} Previously, hyperbranched polymers incorporating PCL segments or MDO have been developed, exhibiting the advantages of high solubility, low viscosity and low crystallinity.^{34–37} However, it is not facile to balance the mechanical strength and degradation. Moreover, it is hard to endow the MDO-based hyperbranched polymer with antifouling abilities, so being a marine antifouling material is difficult.

In the present study, we have prepared a hyperbranched poly(ester-co-vinyl) via a radical ring-opening copolymerization of MDO, VAc and diethylene glycol divinyl ether (DEGDVE) with 2-ethoxythiocarbonylsulfanyl-propionic acid ethyl ester as a chain transfer agent (Scheme 1). The introduction of MDO allows the hyperbranched polymer to degrade. The use of VAc is more facile to optimize the distribution of cleavage units. DEGDVE is used as the branching agent, and its hydrophilicity is favorable to increase the fouling resistance of the coating.

Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China. E-mail: msmcf@scut.edu.cn

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Scheme 1 Schematic illustration of the preparation of a UV curable degradable hyperbranched polymer coating.

The vinyl groups make the hyperbranched polymer form a crosslinked network by UV curing, which further improves its mechanical properties. We examined the degradation and antifouling performances of the hyperbranched polymer coating. Our goal is to develop effective but environmentally friendly marine antifouling systems.

Experimental section

Materials

MDO was synthesized according to previous works.³⁸ Ethyl 2-bromopropionate (EBr, 98%), potassium ethyl xanthate (PEX, 97%), VAc (97%), divinyl adipate (DVA, 99%), and DEGDVE (95%) were all purchased from Sigma-Aldrich Co. Photoinitiator 651 (2,2-dimethoxy-2-phenylacetophenone, DMPA, 99%) was purchased from Sigma-Aldrich Co. 2,2-Azobisisobutyronitrile (AIBN, Aladdin) was recrystallized twice from methanol before use. 1,4-Dioxane was purified by drying over calcium hydride and distilling. Other reagents were used as received. Artificial seawater (ASW) was prepared following ASTM D1141-98 (2003).

Synthesis of 2-ethoxythiocarbonylsulfanyl-propionic acid ethyl ester

2-Ethoxythiocarbonylsulfanyl-propionic acid ethyl ester (EXEP) was synthesized by referring to previous studies,^{39,40} and the detailed synthesis process is described as follows. EBr (1.81 g, 1.0 eq.) and PEX (1.92 g, 1.2 eq.) dissolved in anhydrous aceto-

nitrile (50 mL) were introduced in a one-necked round-bottom flask with a magneton. The reaction was allowed to proceed for 2 h at 0 °C and further for 22 h at room temperature. After filtration and evaporation under reduced pressure, the product was dissolved in dichloromethane. The solution was washed five times with water and then dried over magnesium sulfate. The transparent organic phase was concentrated by evaporation under reduced pressure, and a yellow liquid was obtained. ¹H NMR (600 MHz, CDCl₃), δ(ppm): 4.64 (q, 2H, COOCH₂CH₃), 4.37 (q, 1H, SCH(CH₃)C), 4.2 (q, 2H, SCOCH₂CH₃), 1.57 (d, 3H, SCH(CH₃)C), 1.42 (t, 3H, SCOCH₂CH₃), 1.28 (t, 3H, COOCH₂CH₃). The NMR spectra are shown in Fig. S1.†

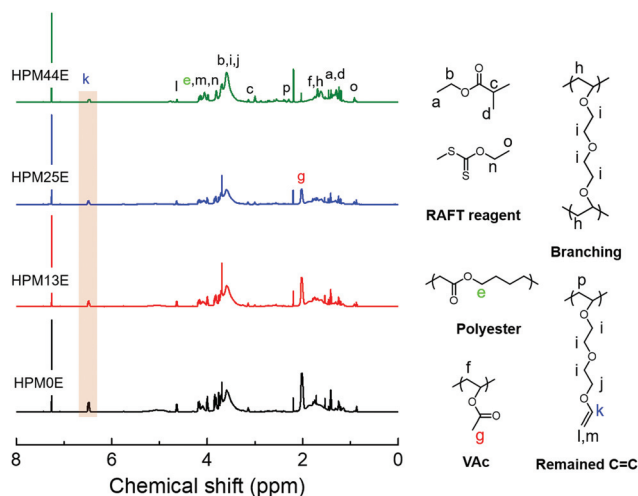
Copolymerization of a UV-curable degradable hyperbranched polymer (DHP)

The copolymerization of MDO, VAc, and DEGDVE was performed under nitrogen in a Schlenk flask using EXEP as the RAFT reagent and AIBN as the initiator (Scheme 1). In a typical experiment, 0.34 g (3 mmol, 30 eq.) of MDO, 0.17 g (2 mmol, 20 eq.) of VAc, 0.79 g (5 mmol, 50 eq.) of DEGDVE, 0.10 g (0.5 mmol, 5.0 eq.) of EXEP, and 16 mg (0.1 mmol, 1.0 eq.) of AIBN were dissolved in 10 mL of 1,4-dioxane and then added to a Schlenk flask with a magnetic stirring bar. After three freezing-pumping cycles, the flask was filled with nitrogen and placed in an oil bath at 70 °C for 24 h. After cooling to room temperature, the mixture was precipitated in *n*-hexane to remove unreacted chemicals. Then, the product was kept under vacuum at 60 °C for 12 h to remove solvents. The obtained hyperbranched polymer is designated as HPMxE

Table 1 Characterization data of hyperbranched polymers

Polymer	Monomer feed mol ratio ^a			Polymer comp. ^b (VAc/MDO/DEGDVE)	M_n ^c (kDa)	D_M ^c	T_g ^d (°C)
	MDO	VAc	DEGDVE				
HPM0E	0	50	50	58/0/42	2.2	2.3	−29.0
HPM13E	10	40	50	43/13/44	2.5	2.6	−31.1
HPM25E	30	20	50	30/25/45	2.4	2.7	−37.7
HPM44E	50	0	50	0/44/56	1.6	2.2	−53.3

^a Monomers/EXEP/AIBN = 100/5/1. ^b Molar percentage determined by ¹H NMR. ^c Determined by GPC. ^d Determined by DSC.

**Fig. 1** ¹H NMR of HPME with different MDO contents in CDCl₃.

(Table 1), where x indicates the molar percentage of MDO determined by ¹H NMR (Fig. 1).

Preparation of a hyperbranched polymer coating

The coating was prepared by the reaction of terminal vinyl groups in the polymer backbone using DMPA (2 wt% with respect to the polymer) as the UV catalyst and THF as the solvent. The solution was coated on a panel and placed in an oven at 60 °C for 2 h to remove THF completely. Then, the panels were subjected to 20 min of UV irradiation, and the hyperbranched polymer coatings were obtained.

Characterization

Proton nuclear magnetic resonance (¹H NMR) spectroscopy was used to analyze the structure and reaction kinetics of the hyperbranched polymer. The number-average molecular weight (M_n) and polydispersity (D_M) were determined by gel permeation chromatography (GPC) on an instrument equipped with refractive index and ultraviolet detectors using poly(ethylene glycol) (PEG, 2.13×10^2 to 21.7×10^5 g mol^{−1}) for calibration. The glass transition temperature was measured on a NETZSCH differential scanning calorimeter (DSC) under a nitrogen flow. The static water contact angle (WCA) was measured on the hyperbranched polymer coating surface using a contact angle system OCA40 (Dataphysics) at 25 °C. A

pull-off adhesion tester (PosiTest AT-A Automatic) was used to test the adhesion strength of the coating on an epoxy panel substrate according to GB/T 5210-2006.

Assessment of degradation properties

The hyperbranched polymer (20 mg) was dissolved in 10 mL of THF and mixed with the same volume of potassium hydroxide/methanol (0.02 M). The mixture was stirred at 30 °C. The solution (1 mL) was taken and subsequently neutralized with hydrochloric acid, followed by evaporation to remove the solvent. The resulting solid was dissolved in dichloromethane and filtered. As the solvent evaporated, the final product was dissolved in THF for GPC determination.

The samples on an epoxy panel (2 × 2 cm²) were prepared via solution-casting with a thickness of ≈200 μm and then cured under 20 min of UV irradiation. The initial weight (W_0) for each dried sample together with its panel were recorded. Then, the samples were immersed in ASW at 25 °C, and removed from ASW at certain intervals and rinsed with de-ionized water. The samples were dried by freeze-drying for 2 days before their dry weights (W_t) were recorded. Different lipase concentrations (0.1, 0.5, 1.0 mg L^{−1}) in ASW were used for enzymatic degradation. The mass loss was calculated using $(W_0 - W_t)/4$. For each polymer, three parallel samples were measured to obtain an average value.

Assessment of antifouling properties

Protein resistance tests. The protein resistance performance was investigated with a quartz crystal microbalance with dissipation (QCM-D) from Q-sense AB (Sweden). The details can be found elsewhere.^{41,42} In brief, the frequency shift (Δf) reflects the change in mass or thickness of the film on the quartz crystal, and the dissipation factor (ΔD) represents the viscoelasticity of the adsorbed layer. The film was prepared by spin-coating on an AT-cut quartz crystal and cured under 20 min of UV irradiation. Afterward, the fibrinogen solution (1.0 mg mL^{−1} in ASW) was delivered to the film surface at a flow rate of 150 μL min^{−1} using ASW as a reference at 25 °C. All the data are from the third overtone ($n = 3$).

Bacterial antiadhesion tests. The marine bacterium *Pseudomonas* sp. was used to evaluate the antibacterial performance of the hyperbranched polymer coating. The coating was prepared on a silicon slide (1 × 1 cm²). The bacterial concentration was spectrophotometrically adjusted to 1×10^7 cells

per mL in ASW. Each sample was immersed in the bacterial suspension for 5 h, washed gently with fresh ASW, and stained with a LIVE/DEAD BacLight Bacterial Viability Kit. The adhered bacterial cells were observed using a fluorescence microscope (Scope A1, Zeiss). The relative bacterial adsorption was analyzed with ImageJ software.

Anti-diatom tests. The diatom *Navicula incerta* was kindly supplied by Xiamen University. The coating was prepared on a silicon slide ($1 \times 1 \text{ cm}^2$). Each sample was immersed in a suspension of diatoms for 12 h of light and 12 h of darkness. Then, it was gently washed with fresh ASW to remove the unattached diatoms and observed with a microscope (Scope A1, Zeiss). Ten random fields of view of each sample were observed to obtain an average value of diatom density.

Results and discussion

The synthesis routes and ^1H NMR spectrum of EXEP are shown in Scheme S1 and Fig. S1,† respectively. The synthetic pathway of the hyperbranched polymer with MDO, VAc and DEGDVE using EXEP as a chain transfer agent is shown in Scheme 1. For convenience, the hyperbranched polymer consisting of 44 mol% MDO is designated as HPM44E, and its conversion of MDO and DEGDVE was examined by ^1H NMR. Fig. S2† shows the time dependence of the monomer conversion during copolymerization. The conversion of MDO is relatively close to that of DEGDVE during the copolymerization, indicating that the reactivity ratios of MDO and DEGDVE are close, and the ester cleavage bonds are randomly distributed in the main chains. The total conversion was about 80% in 24 h, making the final hyperbranched polymer remain with unreacted pendant vinyl groups (Fig. 1) for UV curing. As shown in Fig. 1, we can observe peaks at 6.48 (k), 4.64 (l), and 4.01 ppm (m) assigned to the vinyl group and a peak at 3.60 ppm (i and j) assigned to diethylene glycol units. Peaks at 4.2 (e) and 2.02 ppm (g) are assigned to the protons from MDO and VAc, respectively. The proton signals from the controlling agent EXEP are observed at 4.10 (n), 3.60 (b), 1.41 (a and b), and 0.87 ppm (o). With the molar ratio of DEGDVE, we can estimate the contents of DEGDVE and MDO in the hyperbranched polymer (Table 1).

Eqn (1) is used to determine the degree of branching (DB), and it describes the DB in relation to the number of terminal (T), linear (L), and branching (B) units:^{36,43}

$$\text{DB} = (B + T)/(B + T + L) \times 100\% \quad (1)$$

$$\text{DB} = (B + 2Z)/(B + 2Z + L) \times 100\%. \quad (2)$$

When RAFT polymerization is employed to prepare hyperbranched polymers, the terminal groups originate from the chain transfer agent R-group (R) and Z-group (Z). The numbers of Z-groups and R-groups are approximately equal, so eqn (1) can be modified to eqn (2). By comparing the integrals (Fig. 1b) of the terminal Z-group (Z) protons (o), the DEGDVE (B) protons (i), and the MDO, VAc, and vinyl group (L) protons

(e, g, and k), obtaining the DB for each hyperbranched polymer is possible (Table 2). All hyperbranched polymers with 50% molar feed ratios have a high DB (above 0.3). Actually, since hydrogen transfer occurs on the MDO units, the MDO-based polymer always has a branching structure.⁴⁴ In this work, the DB for this MDO-based hyperbranched polymer might be higher than 0.3. By calculating the integral of vinyl group protons (k), we can derive the molar content of unreacted pendant vinyl groups. As shown in Table 2, the obtained hyperbranched polymers have about 10 mol% of unreacted vinyl groups, making it possible to crosslink by UV curing.

Fig. 2a gives the variation of the number-average molecular weight (M_n) of HPMxE in potassium hydroxide/methanol solution. For HPM0E without ester bonds, M_n slightly decreases from 2200 to 1600 g mol^{-1} after immersing in 0.01 M potassium hydroxide/methanol solution for 5 h. This is because VAc in the polymer chain hydrolyzes under alkaline conditions, leading to the reduction of molecular weight. For HPM13E with 13 mol% MDO, M_n sharply decreases from 2500 to 600 g mol^{-1} and levels off, indicating that the ester linkages are broken and the polymer degrades into small fragments. For HPMxE with higher MDO content (25 and 44 mol%), M_n decreases more sharply than that for HPM13E, indicating that they have faster degradation rates. Notably, the M_n of the degraded fragments decreases with MDO content. As discussed above, the small difference in reactivity ratios between MDO, VAc and DEGDVE allows MDO to be randomly distributed along the main chain. Therefore, this hyperbranched polymer possesses a constant degradation rate with much smaller degraded fragments, which is favorable for improving the antifouling ability of the coating and environmental protection.

Hyperbranched polymers usually exhibit high solubility, low melt viscosity, low crystallinity, and high end-up functional group density.²⁹ Yet, MDO-based hyperbranched polymers have a such low glass transition temperature (T_g) that it is difficult for them to form a film. Thanks to the unreacted vinyl groups in the hyperbranched polymer in this study, it can be crosslinked under UV irradiation. Although each uncured hyperbranched polymer has a T_g below -29°C (Table 1), the coating consisting of the cured hyperbranched polymer can readily form a film. The adhesion strength of the HPMxE coating to the epoxy panel decreases from 1.3 to 0.5 MPa as the MDO content increases (Fig. S3†). This is understandable because the crosslinking reduces the segment mobility, and

Table 2 DB and unreacted vinyl content of hyperbranched polymers calculated from NMR results

Polymer	DB	Vinyl (mol%)
HPM0E	0.27	10.60
HPM13E	0.31	9.30
HPM25E	0.31	11.12
HPM44E	0.44	8.80

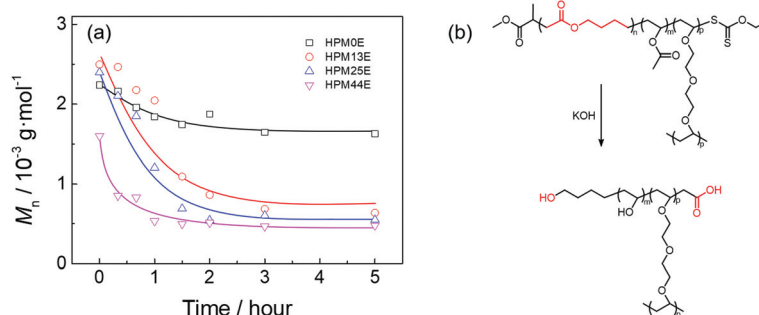


Fig. 2 Degradability of HPMxE. (a) Variation of the molar mass of HPMxE degraded in potassium hydroxide/methanol solution at 25 °C. (b) Schematic of degradation and hydrolysis of HPMxE.

the presence of VAc units further hinder the mobility of the chain. Thus, the mechanical properties improve.

The hyperbranched polymer can degrade into small fragments with M_n less than 650 g mol^{-1} in 0.01 M potassium hydroxide/methanol solution. We also investigated the mass loss of the hyperbranched polymer coating in ASW (Fig. 3a). For the HPM0E coating without MDO, the value of mass loss is 0.4 mg cm^{-2} after 14 days. A portion of the observed change in the mass loss curve of this sample could be attributed to the dissolution of small molecules and hydrolysis of VAc. The VAc unit in the polymer chain can hydrolyze to hydroxyl and carboxylic acid under alkaline conditions as well as high humidity conditions.^{45,46} This can be further confirmed by ATR-IR (Fig. S4[†]); the absorption peaks at 3476 cm^{-1} (ν_{OH}) and 1647 cm^{-1} (δ_{OH}) in the spectrum of HPM0E indicate the hydrolysis of VAc. Therefore, a relatively obvious change in mass occurs after immersion in weakly alkaline seawater (pH = 8.2) for 14 days. For an HPM13E coating containing 13 mol% MDO, the mass loss shows a first-order linear relationship with immersion time, and its slope is larger than that for HPM0E, indicating that the degradation rate is faster than that for HPM0E. Thanks to the hydrolysis of VAc and the degradation of the ester linkage, the separated small fragments would rapidly dissolve in water (Fig. 2b), leading to a reduction of coating mass. Further increasing the MDO content (25 and 44 mol%) can increase the degradation rate, indicating that

the contribution to mass loss from the degradation of MDO is more than that from the hydrolysis of VAc. Nevertheless, this hyperbranched polymer coating exhibits a fast and controllable degradation rate, which is favorable for its antifouling.

Many enzymes are produced by fouling organisms in the marine environment that can speed up the fracture of the ester linkage.¹⁰ For simplicity, we measured the time dependence of the mass loss of the HPM44E coating in different lipase PS concentrations (0, 0.1, 0.5, and 1.0 mg mL^{-1}) (Fig. 3b). The coating exhibits an increasing degradation rate with the increase of lipase PS in ASW with a mass change above 1.0 mg cm^{-2} after 14 days. The facts indicate that the hyperbranched polymer coating is enzyme responsive.

Polyethylene glycol containing ethylene glycol units exhibits resistance to protein, marine bacteria, and diatoms due to its high hydrophilicity.^{47–50} The hyperbranched polymer coatings here have about 50 mol% diethylene glycol units, so they are expected to have good antifouling ability. To examine their antifouling performance, we prepared a control coating with DVA instead of DEGDE (Scheme S2[†]) designated as HPM0D. As shown in Fig. 4, we investigated the adsorption of fibrinogen on HPM0D and HPMxE coatings in ASW at 25 °C using QCM-D to evaluate the protein resistance. For HPM0D without any diethylene glycol units, Δf rapidly decreases after the introduction of fibrinogen and then gradually levels off. A marked decrease in Δf and increase in ΔD relative to the baseline after

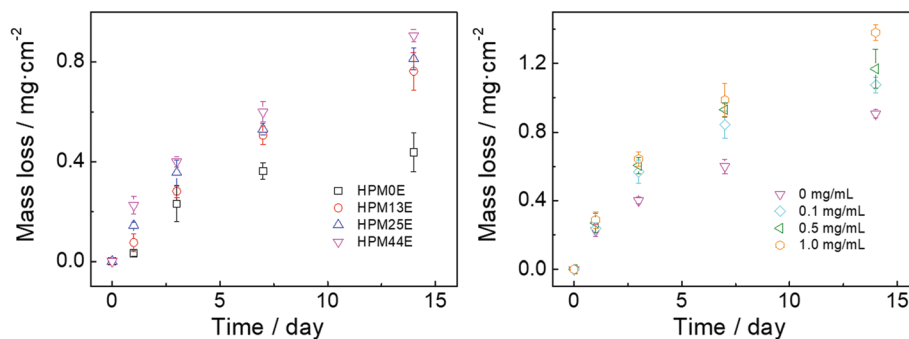


Fig. 3 (a) Time dependence of the mass loss of HPMxE coatings in ASW and (b) variation of the mass loss of the HPM44E coating in different concentrations of lipase PS in ASW at 25 °C.

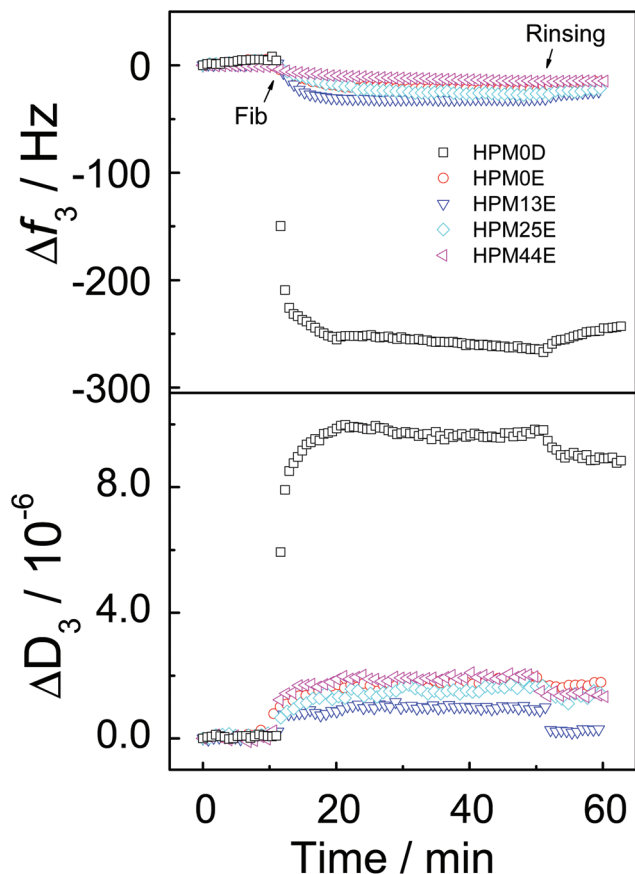


Fig. 4 Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on HPM0D and HPMxE coatings in ASW at 25 °C.

rinsing with ASW is found, indicating that the mass increases due to the absorption of fibrinogen. For HPM0E with about 43 mol% diethylene glycol units, a smaller decrease in Δf and increase in ΔD relative to the baseline indicate that the mass increase caused by the absorption of fibrinogen is smaller

than that for HPM0D. As the MDO content increases, the decrease in Δf for HPM13E, HPM25E and HPM44E coatings changes a little, indicating that the absorption mass of fibrinogen slightly decreases. This is probably because the T_g of HPMxE decreases with MDO content, leading to an increase in the hydration capacity of the coating surface.

This can be further confirmed by WCA tests (Fig. S5†); the WCA of the HPM0D coating is about 80°, whereas those of HPMxE coatings are all about 60°, indicating that HPMxE coatings are more hydrophilic than HPM0D. The WCA of HPMxE coating decreases from 62° to 55° as the MDO content increases from 0 to 44 mol%, indicating the increase in hydration. In short, the HPMxE coatings exhibit excellent protein resistance performance, and the hyperbranched polymer coating without any biocidal groups is potentially used in marine antifouling.

Fig. 5 shows fluorescence microscopy photographs of the HPMxE coating surfaces after immersion in a suspension of the marine bacterium *Pseudomonas* sp.. For the silicon surface, many green fluorescent spots were observed, indicating that a large number of marine bacteria adhered onto the silicon surface. The same phenomenon was observed on the HPM0D coating surface, indicating that the HPM0D coating surface without ethylene glycol units could not inhibit bacterial adhesion. For HPM0E with 42 mol% diethylene glycol units, a few green fluorescent spots were observed, that is, a few bacteria adhered onto the HPM0E coating surface. Particularly, almost no bacteria adhered onto the HPM13E, HPM25E and HPM44E surfaces. These facts clearly indicate that the diethylene glycol units play a critical role in inhibiting bacterial adhesion. In other words, the hyperbranched polymer coating with diethylene glycol units can effectively prevent marine bacteria from the surface.

The surface hydrophilicity can also affect the adhesion of diatoms. We measured the anti-diatom ability of HPMxE coatings. Fig. 6 shows photographs and density of the diatoms on HPM0D and HPMxE coating surfaces after 24 h of immersion

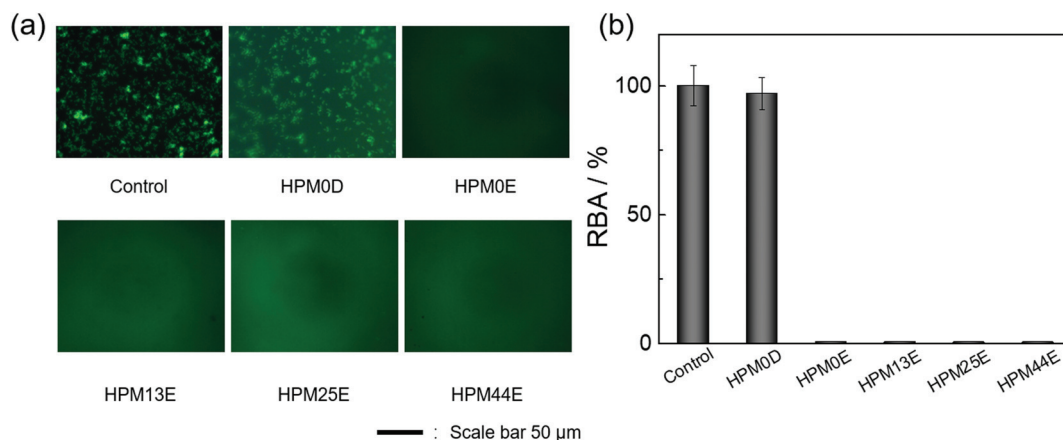


Fig. 5 Fluorescence microscopy photographs (a) and relative bacterial adsorption (RBA) (b) of *Pseudomonas* sp. adhered on HPMxE coating surfaces.

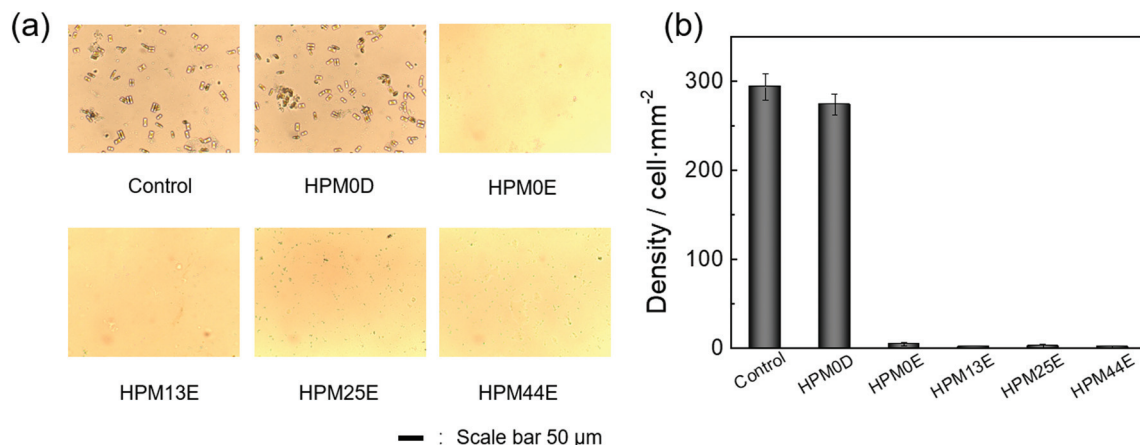


Fig. 6 Photographs (a) and density (b) of the diatoms on HPMxE coating surfaces after 24 h of immersion in a *Navicula incerta* suspension.

in *Navicula incerta* suspension. For silicon and HPM0D without any ethylene glycol units, a dozen diatoms are observed in the photograph, and their density is ≈ 300 cells per square millimeter (Fig. 6b), showing that the HPM0D coating surface does not affect the adhesion of diatoms. For HPM0E, we can observe a very clean surface, indicating that the coating containing 42 mol% ethylene glycol units exhibits a remarkable performance for inhibiting the adhesion of diatoms. This similar result can be obtained on HPM13E, HPM25E and HPM44E surfaces. The ethylene glycol units can be readily hydrated forming a water-based protective film, making the attaching of the diatoms impossible. These facts further indicate that the hyperbranched polymer coatings with ethylene glycol units have good antifouling ability.

Conclusions

We have prepared UV-curable hyperbranched poly(ester-co-vinyl) *via* radical ring-opening copolymerization of 2-methylene-1,3-dioxepane (MDO), vinyl acetate (VAc) and diethylene glycol divinyl ether (DEGDVE). Such a hyperbranched polymer exhibits tunable degradation and excellent antifouling performance. Since MDO and the divinyl monomer have a close reactivity ratio, they are randomly distributed in the hyperbranched polymer chain, and the polymer can degrade into small fragments. The degradation rate of the polymer can be well regulated by the MDO content. The presence of diethylene glycol units allows the hyperbranched polymer to exhibit protein resistance, antibacterial ability, and anti-diatom activity. The hyperbranched polymer coating is expected to be used in marine antifouling.

Author contributions

G. Z. Zhang and C. F. Ma conceived the idea and designed the experiments. X. Q. Ai carried out the experiments, interpreted the results and prepared the manuscript. J. S. Pan and

Q. Y. Xie reviewed and edited the manuscript. All authors have given approval to the final version of the manuscript.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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